



KTU NOTES

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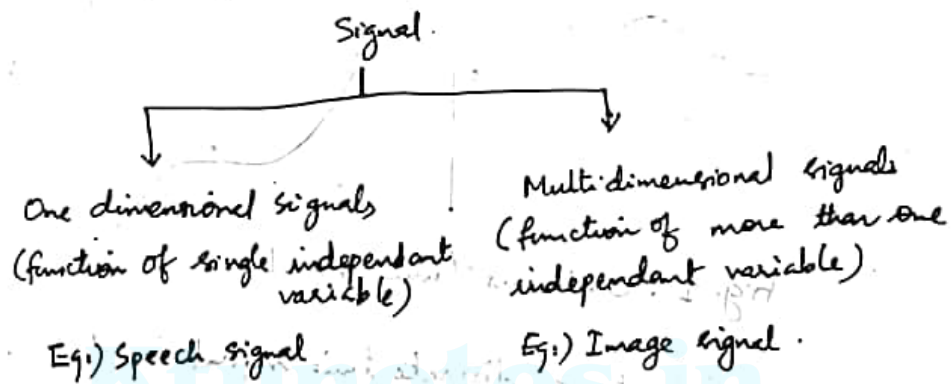
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Signal · MODULE I

A signal is formally defined as a function of one or more variables that conveys information on the nature of a physical phenomenon.

Eg: speech signal, music signal, image signals, Radar signal, ECG signal, EEG signal etc.



System

A system is an entity that manipulates one or more signals to accomplish a function, thereby yielding new signals.



Eg: Automatic speaker recognition system, Communication system, control systems, Remote sensing, biomedical signal processing system etc.

Classification of Signals:

1. Continuous time and discrete time signals:

A signal $x(t)$ is said to be continuous time signal if it is defined for all time t .

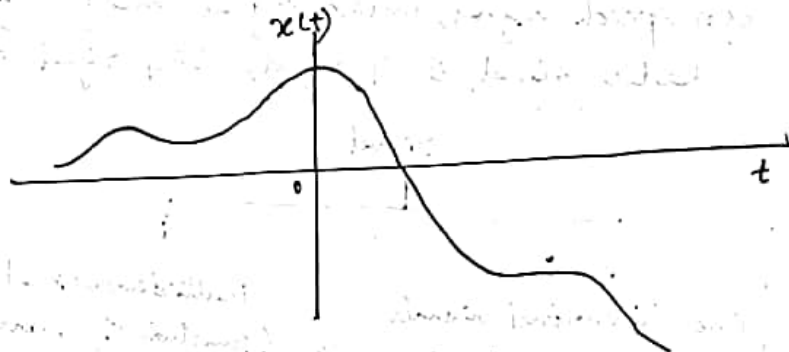


Fig. 1: Continuous time signal

In contrast, a discrete time signal is defined only at discrete instants of time. Thus, the independent variable has discrete values only, which are uniformly spaced.

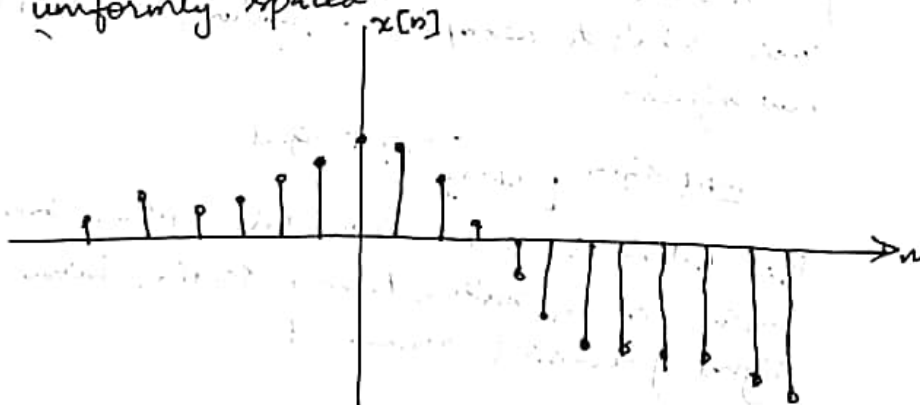


Fig. 2: Representation of $x(t)$ as a discrete time signal $x[n]$.

A discrete time signal is often derived from continuous signal by sampling. Let T_s denote the sampling period and n denote an integer,

Then, $x[n] = x[nT_s]$, $n = 0, \pm 1, \pm 2, \dots$

2. Even and odd signals

$x(t)$ is said to be even signal if,

$$x(-t) = x(t) \quad , \text{symmetric about vertical axis}$$

$x(t)$ is said to be an odd signal if,

$$x(-t) = -x(t) \quad , \text{antisymmetric about time origin}$$

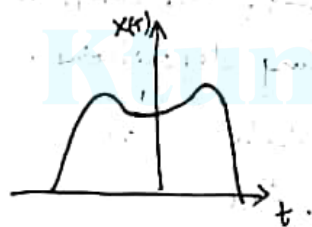


Fig.3 Even signal

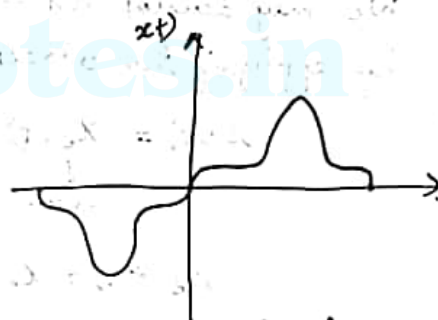


Fig.4 odd signal.

Q1. Consider the signal,

$$x(t) = \begin{cases} \sin\left(\frac{\pi t}{T}\right) & , -T \leq t \leq T \\ 0 & , \text{otherwise} \end{cases}$$

Is the signal $x(t)$ an even or an odd function of time t ?

Ans: Replacing t with $-t$

$$x(-t) = \begin{cases} \sin\left(-\frac{\pi t}{T}\right), & -T \leq t \leq T \\ 0, & \text{otherwise.} \end{cases}$$

$$= \begin{cases} -\sin\left(\frac{\pi t}{T}\right), & -T \leq t \leq T \\ 0, & \text{otherwise.} \end{cases}$$

$$= -x(t) \text{ for all } t.$$

Hence $x(t)$ is an odd signal.

We may develop an even-odd decomposition of $x(t)$ by applying the corresponding definitions.

$$x(t) = x_e(t) + x_o(t) \quad \text{--- (1)}$$

$\downarrow$ even $\downarrow$ odd.

$$x_e(-t) = x_e(t)$$

$$x_o(-t) = -x_o(t)$$

Putting $t = -t$,

$$x(-t) = x_e(-t) + x_o(-t)$$

$$= x_e(t) - x_o(t) \quad \text{--- (2)}$$

$$\textcircled{1} + \textcircled{2} \quad x(t) + x(-t) = 2x_e(t)$$

$$x_e(t) = \frac{1}{2} (x(t) + x(-t))$$

$$① - ② \Rightarrow x(t) - x(-t) = 2 x_o(t)$$

$$x_o(t) = \frac{x(t) - x(-t)}{2}$$

Q Find the even and odd components of the signal

$$x(t) = e^{-2t} \cos t$$

$$x(-t) = e^{2t} \cos(-t) \\ = e^{2t} \cos t$$

$$x_e(t) = \frac{1}{2} (x(t) + x(-t))$$

$$= \frac{1}{2} (e^{-2t} \cos t + e^{2t} \cos t)$$

$$= \cos t \frac{e^{-2t} + e^{2t}}{2} \quad \left| \frac{e^x + e^{-x}}{2} = \cosh(x) \right.$$

$$= \cosh(2t) \cos t$$

$$x_o(t) = \frac{1}{2} (e^{-2t} \cos t - e^{2t} \cos t)$$

$$= \cos t \frac{(e^{-2t} - e^{2t})}{2}$$

$$= -\cos t \frac{(e^{2t} - e^{-2t})}{2} \quad \left| \frac{e^x - e^{-x}}{2} = \sinh(x) \right.$$

$$= -\sinh(2t) \cos t$$

3. Periodic signals and nonperiodical signals.

A periodic signal $x(t)$ is a function of time that satisfies the condition

$$x(t) = x(t+T) \quad \text{for all } t,$$

where T is a positive constant.

The smallest value of T that satisfies the above equation is called the fundamental period of $x(t)$.

The fundamental frequency f is,

$$f = \frac{1}{T}$$

The angular frequency is,

$$\omega = 2\pi f = \frac{2\pi}{T}, \quad \text{since there are } 2\pi \text{ radians in one cycle.}$$

Ex: $x(t)$.

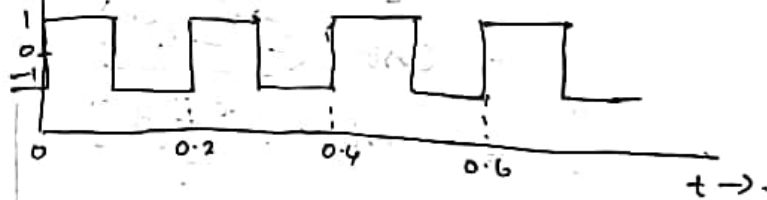


Fig. 5: Square wave with amplitude $A=1$ and period $T=0.25$

4. Deterministic signals and random signals.

A deterministic signal about which there is no uncertainty with respect to its value at any time. Eg:) Fig. 5.

A random signal is a signal about which there is uncertainty before it occurs. Such a signal may be viewed as belonging to an ensemble, or a group, of signals, with each signal in the ensemble having a different waveform. Moreover each signal within the ensemble has a certain probability of occurrence. The ensemble of signals is referred to as a random process.

Eg:) Signal received in a radio communication system.

EEG (Electroencephalogram) signal.

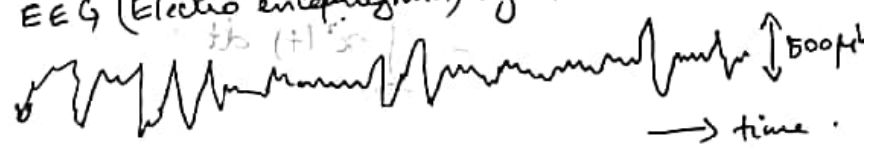


Fig. 6: Single channel EEG.

EEG is measured as the voltage between two electrodes placed over the head.

$$V(t) = \frac{1}{T} \int_{-T}^{+T} V(t) dt = 9$$

5. Energy signals and power signals

Consider a voltage $v(t)$ developed across a resistor R , producing a current $i(t)$. Then,

$$p(t) = i^2(t) R = \frac{v^2(t)}{R}$$

For a resistance of 1Ω ,

$$p(t) = i^2(t) = v^2(t)$$

The instantaneous power of the signal is,

$$p(t) = x^2(t)$$

The total energy,

$$E = \lim_{T \rightarrow \infty} \int_{-T/2}^{T/2} x^2(t) dt$$

$$= \int_{-\infty}^{\infty} x^2(t) dt$$

Average power,

$$P = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x^2(t) dt$$

The time averaged power of a periodic signal $x(t)$ of fundamental period T is,

$$P = \frac{1}{T} \int_{-T/2}^{T/2} x^2(t) dt$$

In the case of discrete time signal $x[n]$,

The total energy of $x[n]$ is,

$$E = \sum_{n=-\infty}^{\infty} x^2[n].$$

Its average power is,

$$P = \lim_{N \rightarrow \infty} \frac{1}{2N} \sum_{n=-N}^N x^2[n].$$

Average power in a periodic signal $x[n]$ with fundamental period N is given by,

$$P = \frac{1}{N} \sum_{n=0}^{N-1} x^2[n].$$

A signal is referred to as an energy signal if and only if the total energy of the signal satisfies the condition,

$$0 < E < \infty, \text{ For energy signal, } P = 0.$$

A signal is referred to as power signal if and only if the average power of the signal satisfies the condition,

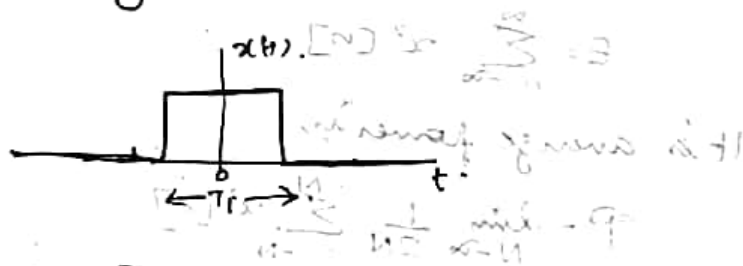
$$0 < P < \infty$$

For power signal, $E = \infty$.

Periodic signals & random signals are usually viewed as power signals.

Both deterministic and nonperiodic are usually viewed as energy signals.

Qn) What is the total energy of the rectangular pulse shown in figure.



$$E = \int_{-\infty}^{\infty} x^2(t) dt$$

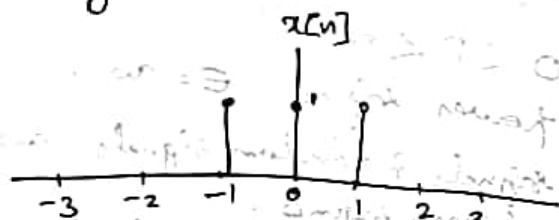
$$= \int_{-T/2}^{T/2} A^2 dt$$

$$= A^2 \left[\frac{t}{1} \right]_{-T/2}^{T/2}$$

$$= A^2 \left[\frac{T}{2} - \left(-\frac{T}{2}\right) \right]$$

$$= A^2 T$$

Qn) Determine the total energy of the discrete time signal



$$E = \sum_{n=-\infty}^{\infty} x^2[n]$$

$$= \sum_{n=-1}^1 x^2[n]$$

$$= 1 + 1 + 1 = \underline{\underline{3}}$$

Basic operations on signals:

Operations performed on dependant variable:

i) Amplitude scaling:

$$y(t) = c x(t).$$

Eg: Amplifier, resistor.

ii) Addition:

$$y(t) = x_1(t) + x_2(t)$$

Eg: audio mixer.

iii) Multiplication:

$$y(t) = x_1(t) x_2(t)$$

Eg: AM radio signal.

iv) Differentiation:

$$y(t) = \frac{d}{dt} x(t).$$

Eg: Inductor.

v) Integration:

$$y(t) = \int_{-\infty}^t x(\tau) d\tau.$$

Eg: Capacitor.

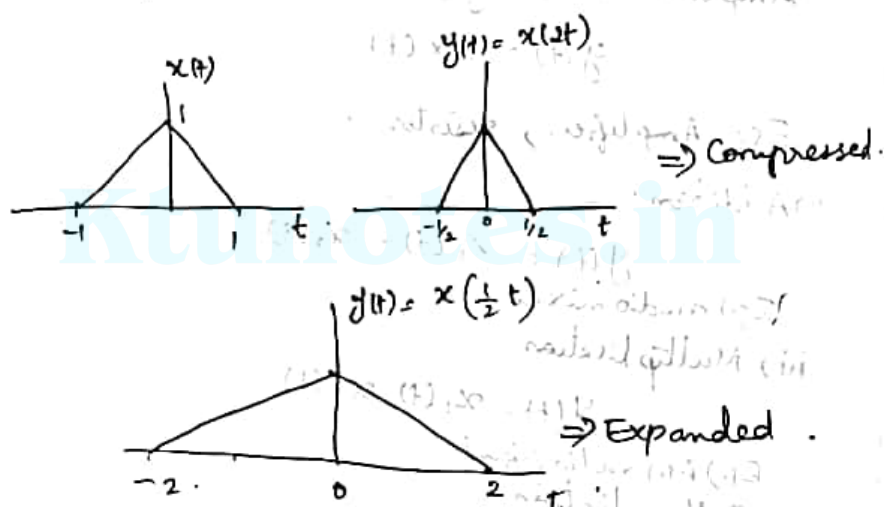
Operations Performed on the independent Variable

Time Scaling:

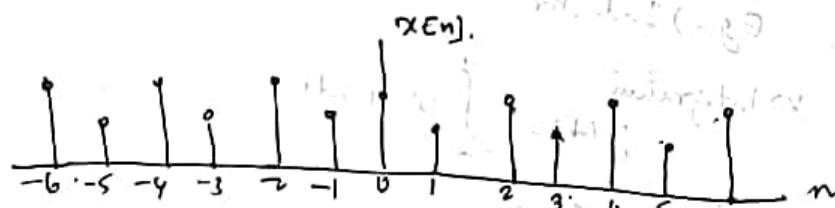
$$y(t) = x(at)$$

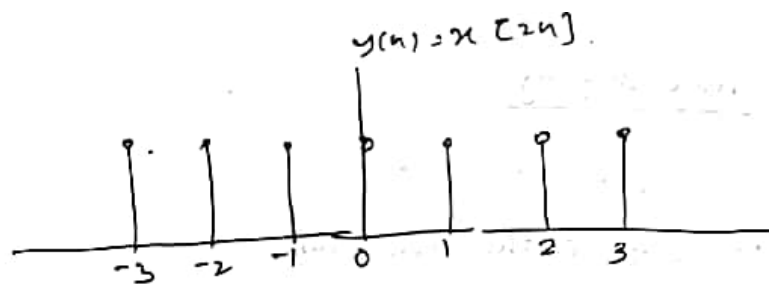
If $a > 1$, $y(t)$ is a compressed version.

If $0 < a < 1$, $y(t)$ is an expanded version.



Discrete time signal





compressed by a factor of 2.

Qn Let $x[n] = \begin{cases} n, & \text{for } n \text{ odd} \\ 0, & \text{otherwise} \end{cases}$

Determine $y[n] = x[2n]$.

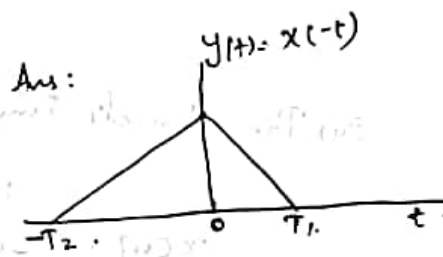
Ans: $y[n] = 0$, for all n .

Reflection (Time Reversal)

$$y(t) = x(-t)$$

$y(t)$ represents a reflected version of $x(t)$ about $t=0$.

Qn) Find the reflected version of $x(t)$.



Reflected version.

Time shifting

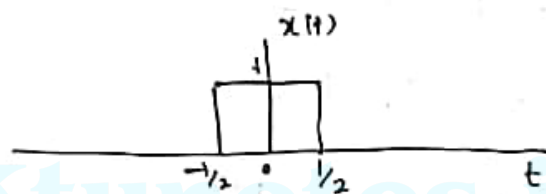
$$y(t) = x(t - t_0)$$

where t_0 is the time shift.

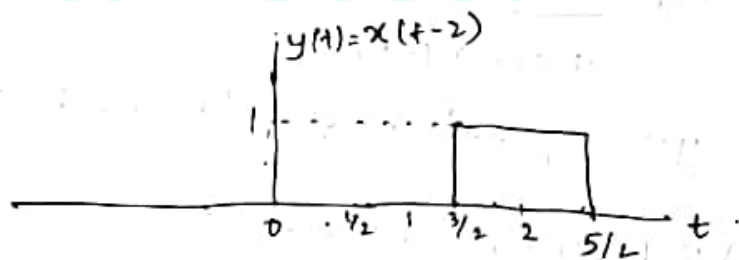
If $t_0 > 0$, shifted towards right.

If $t_0 < 0$, " " left.

Ex) Figure shows a rectangular pulse $x(t)$ of unit amplitude and unit duration. Find $y(t) = x(t-2)$



Ans:



Ex) The discrete time signal

$$x[n] = \begin{cases} 1, & n = 1, 2 \\ -1, & n = -1, -2 \\ 0, & n = 0, \text{ and } |n| > 2 \end{cases}$$

Find $y[n] = x[n+3]$.

for

Ans: $t_0 = -3$

$$y[n] = \begin{cases} 1, & n = -2, -1 \\ -1, & n = -4, -5 \\ 0, & n = -3, n < -5, n > -1 \end{cases}$$

$$\begin{cases} |n| > 2 \\ n > 2 \\ n < -2 \\ \therefore \begin{cases} n > (2-3) \\ n < (-2-3) \end{cases} \end{cases}$$

Precedence rule for time shifting and time scaling.

$$y(t) = x(at-b)$$

First, the time shifting operation is to be performed.

$$v(t) = x(t-b)$$

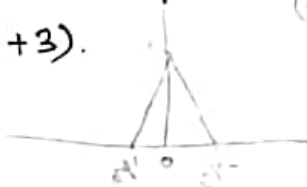
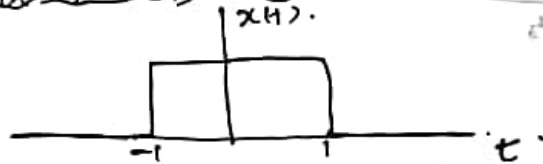
Next, the time scaling operation is performed on $v(t)$

$$y(t) = v(at)$$

$$= x(at-b)$$

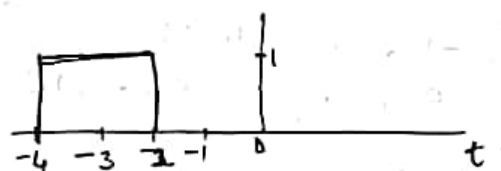
Qn) Consider the rectangular pulse $x(t)$ of unit amplitude and a duration of 2 time units depicted in the figure. Find $y(t) = x(2t+3)$.

Ans: $a=2$ $b=-3$

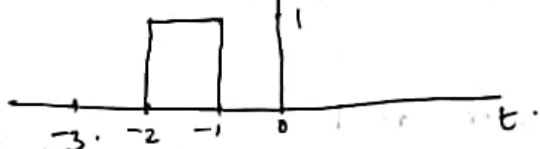


Ans: $a=2$ $b=-3$

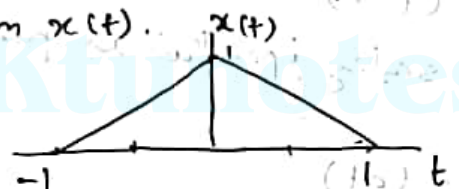
$$v(t) = x(t+3)$$



$$y(t) = v(2t)$$



Qn) A triangular pulse signal $x(t)$ is depicted in Figure. Sketch each of the following signals derived from $x(t)$.



a) $x(3t)$

d) $x(2(t+2))$

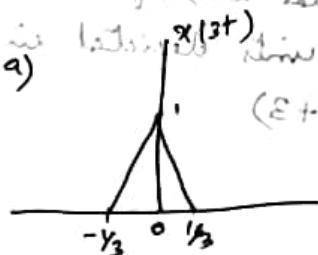
b) $x(3t+2)$

e) $x(2(t-2))$

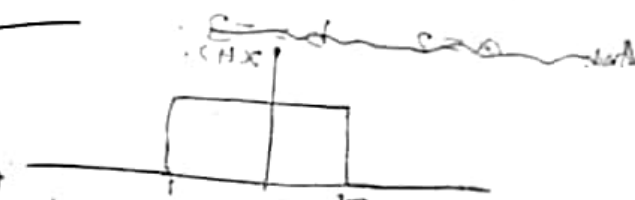
c) $x(-2t-1)$

f) $x(3t) + x(3t+2)$

Ans: a)



(or) Consider the rectangular pulse signal $x(t)$ as shown in Figure. Find $y(t) = x(t+3)$ and $z(t) = x(2t)$.



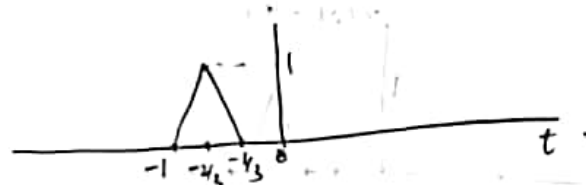
b) $x(3t+2)$.

$v = x(t+2)$

$x = x(t+2)$

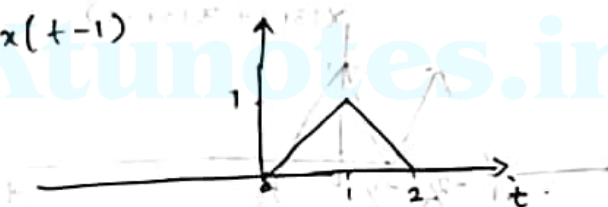


$y = v(3t) = x(3t+2)$

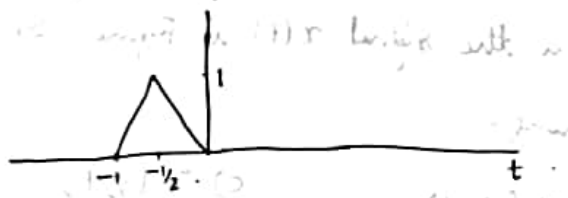


c) $x(-2t-1)$

$x = x(t-1)$



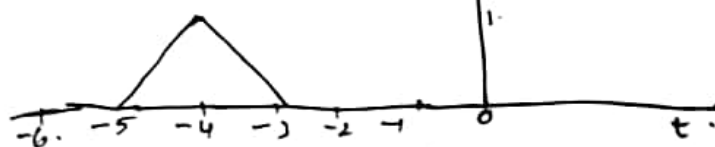
$y = v(-2t) = x(-2t-1)$



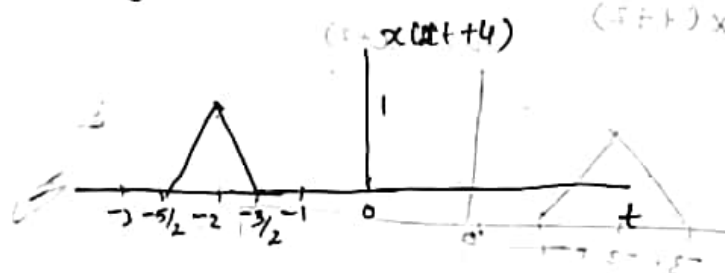
d) $x(2(t+2)) = x(2t+4)$

$v = x(t+4)$

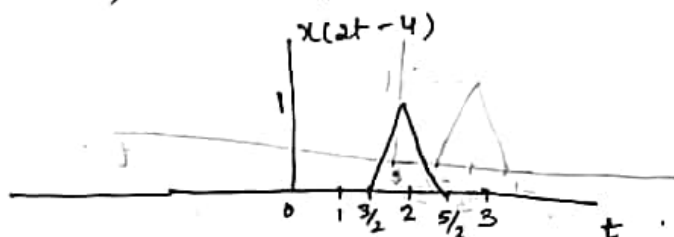
$x(t+4)$



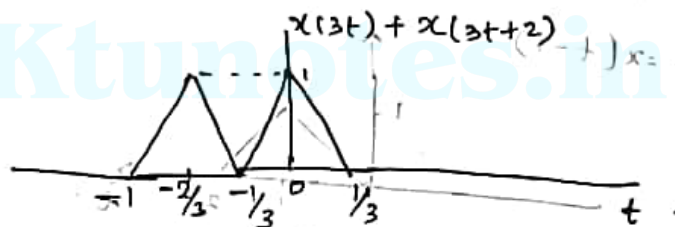
$$y = v(2t) = x(x^2 + 4). \quad (-5 + 7.5)x. \quad (d)$$



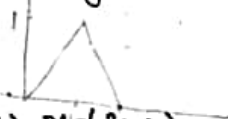
c) $x(2t-2) = x(2t-4)$ $(+1)^2 x = (18) x = 18$



f).



(9) Given the signal $x(t)$ in Figure. Sketch the following.

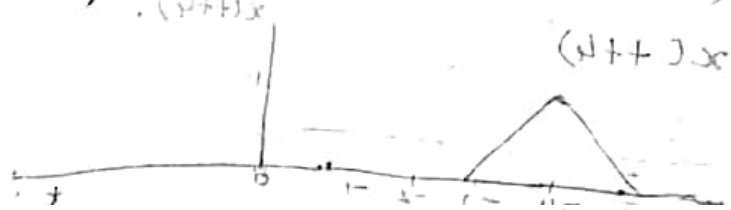


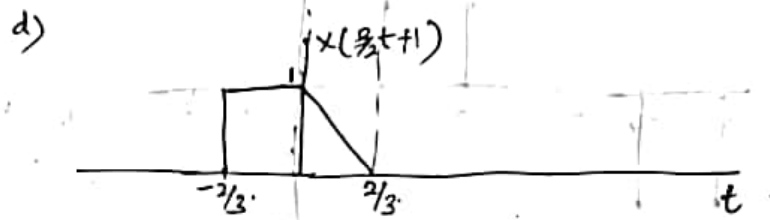
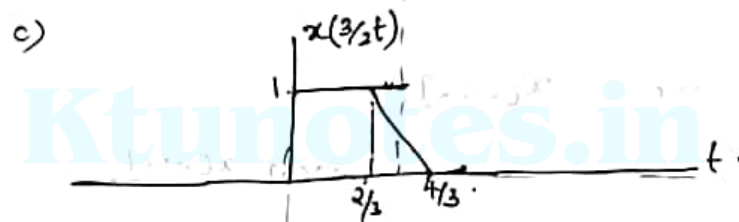
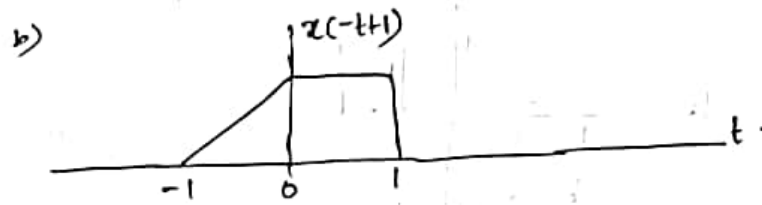
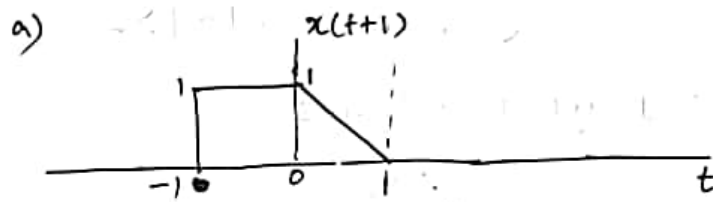
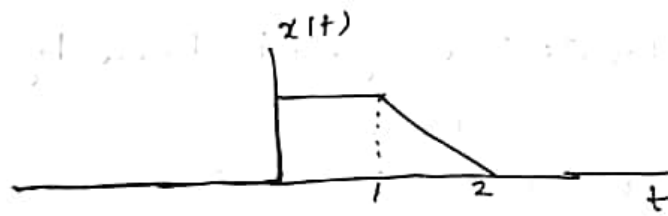
$$a) x(t+1)$$

c) $x - (3/2 \cdot t)$

b) $x(-t+1)$

$$(1 + \frac{1}{2}d) x (\frac{3}{2}t + 1)$$

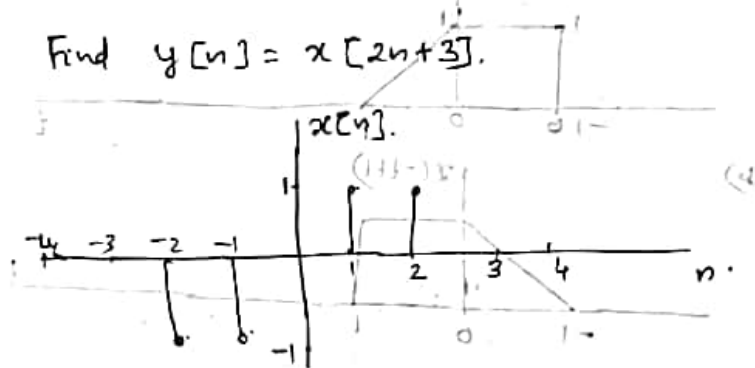




Q. A discrete time signal is defined by,

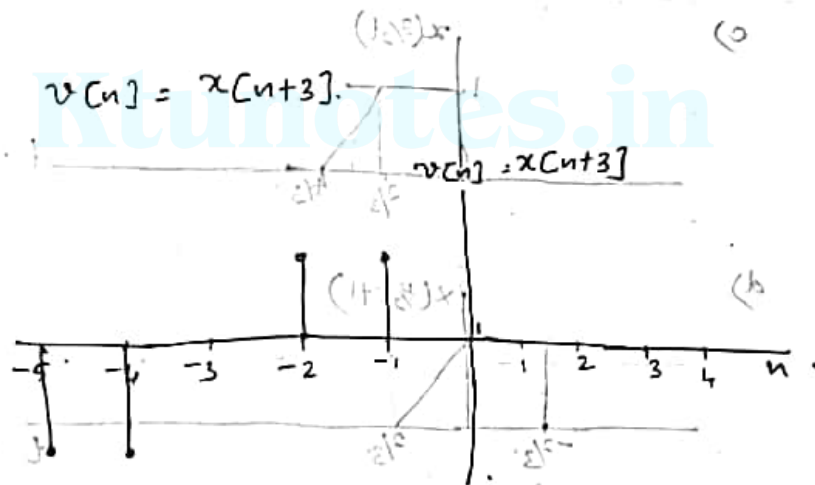
$$x[n] = \begin{cases} 1, & n=1, 2 \\ -1, & n=-1, -2 \\ 0, & n=0 \text{ and } |n| > 2 \end{cases}$$

Find $y[n] = x[2n+3]$.

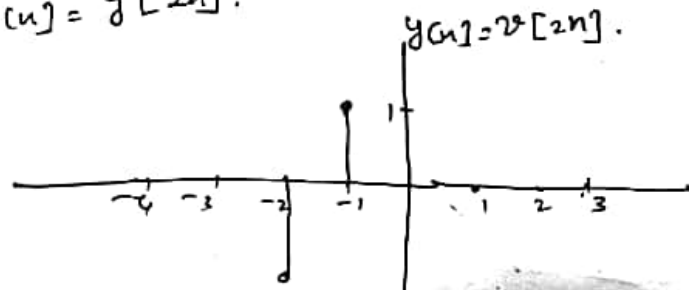


Ans:

$$v[n] = x[n+3]$$

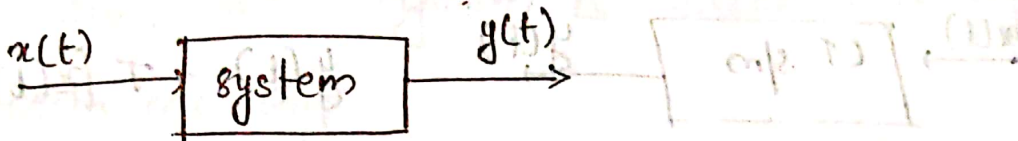


$$y[n] = v[2n]$$



System

A system is defined as a physical device that generates a response or output signal for a given input signal.

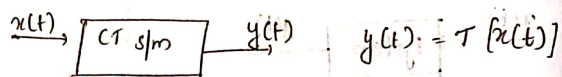


Classification of systems

1. Continuous time and discrete time system.
2. Causal and non causal system.
3. Linear and Non linear system.
4. Time variant and time invariant system.
5. Stable and unstable system.
6. Static and dynamic systems.

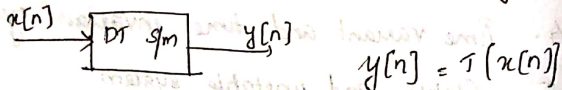
→ Continuous time and discrete time system

A continuous time system is one which operates on a continuous signal and produces a continuous-time output signal



Eg: Amplifiers, filters, motors, etc

A discrete time system is one which operates on a discrete time signal and produces a discrete time output signal



Static and dynamic system

o/p depends on the input at that instant but on not on past or future values of input. → Static s/m or memoryless

Otherwise, the system is said to be dynamic or with memory.

$$\left. \begin{aligned} y(t) &= x^2(t) \\ y(n) &= n x(n) \end{aligned} \right\} \text{static}$$

$$\left. \begin{aligned} y(t) &= \frac{d}{dt} x(t) \\ y(n) &= x(n-1) \end{aligned} \right\} \text{dynamic}$$

→ Find whether or not the following systems are dynamic

1) $y(t) = x(t-2)$

2) $y(n) = x(n+2)$

3) $y(t) = x^2(t)$

A. $y(t) = x(t-2)$
 $t=0 \rightarrow y(0) = x(-2) \rightarrow$ depend on past value
 $\therefore$ Dynamic s/m.

$$2. y(n) = x(n+2)$$

$$y(0) = x(2) \rightarrow \text{depend on future}$$

$\therefore$ Dynamic

$$3. y(t) = x^2(t)$$

$$y(0) = x^2(0) \rightarrow \text{static}$$

$\rightarrow$ Causal and non causal systems

A causal system is one for which the output at any time t depends on the present and past inputs but not future inputs.

A non causal s/m is one whose output depends on future values.

$$\left. \begin{array}{l} y(t) = x(t) + x(t-1) \\ y(n) = nx(n) + x(n-3) \end{array} \right\} \text{causal}$$

$$\left. \begin{array}{l} y(t) = x(t+3) + x^2(t) \\ y(n) = x(2n) \end{array} \right\} \text{non causal.}$$

$\rightarrow$ Check whether the following s/ms are causal or not.

$$1) y(n) = x(n) + \frac{1}{x(n-1)} \quad c$$

$$2) y(t) = x^2(t) + x(t-2) \quad c$$

$$3) y(t) = x(t-2) + x(2-t) \quad \text{NC}$$

$$4) y(t) = \int_{-\infty}^{2t} x(\tau) d\tau \quad \text{NC}$$

$$5) y(n) = x(-n) \quad \text{NC}$$

$$6) y(n) = x(n^2) \quad \text{NC}$$

$$1. y(n) = x(n) + \frac{1}{x(n-1)}$$

$$n=0 \quad y(0) = x(0) + \frac{1}{x(-1)}$$

$$n=1 \quad y(1) = x(1) + \frac{1}{x(0)}$$

$$n=-1 \quad y(-1) = x(-1) + \frac{1}{x(-2)} \quad \text{past & present causal}$$

$$3. y(t) = x(t-2) + x(2-t)$$

$$t=0 \quad y(0) = x(-2) + x(2) \rightarrow \text{Non-causal}$$

past future

$$4. y(t) = \int_{-\infty}^{2t} x(\tau) d\tau \quad \int x(\tau) d\tau = z(\tau)$$

$$t=0 \quad y(0) = \int_{-\infty}^0 x(\tau) d\tau = z(0) - z(-\infty)$$

present past

$$t=1 \quad y(1) = \int_{-\infty}^2 x(\tau) d\tau = z(2) - z(-\infty)$$

future past

Non-causal.

$$5. y(n) = x[-n]$$

$$y(0) = x(0) \rightarrow \text{present}$$

$$y(1) = x(-1) \rightarrow \text{past}$$

$$y(-1) = x(1) \rightarrow \text{future}$$

Non-causal

$$6. y(n) = x(n^2)$$

$$y(0) = x(0)$$

$$y(1) = x(1)$$

$$y(2) = x(4) \rightarrow \text{future}$$

Non-causal.

→ Linear and non-linear system

A system that satisfies the superposition & homogeneity principle is said to be a linear system

Superposition principle states that the

$$\text{If } x_1(t) \rightarrow y_1(t)$$

$$x_2(t) \rightarrow y_2(t), \text{ then s/m is linear if}$$

the i/p $a x_1(t) + b x_2(t)$ produces an

o/p $a y_1(t) + b y_2(t)$, where a & b are const

Superposition.

$$x_1(t) + x_2(t) \rightarrow y_1(t) + y_2(t)$$

homogeneity.

$$x(t) \rightarrow y(t)$$

$$a x_1(t) \rightarrow a y_1(t)$$

$$b x_2(t) \rightarrow b y_2(t)$$

$$\text{Linear} \rightarrow a x_1(t) + b x_2(t) \rightarrow a y_1(t) + b y_2(t)$$

→ Check whether the following s/m are linear or not.

$$1. \frac{dy}{dt} + 3t y(t) = t^2 x(t).$$

$$\text{for i/p } x_1(t) \rightarrow \text{o/p } y_1(t).$$

$$\frac{dy_1(t)}{dt} + 3t y_1(t) = t^2 x_1(t) \quad \text{--- ①}$$

$$x_2(t) \rightarrow y_2(t).$$

$$\frac{dy_2(t)}{dt} + 3t y_2(t) = t^2 x_2(t) \quad \text{--- ②}$$

$$① \times a + ② \times b$$

$$a \frac{dy_1(t)}{dt} + 3at y_1(t) + b \frac{dy_2(t)}{dt} + 3bt y_2(t) = at^2 x_1(t) + bt^2 x_2(t)$$

$$\frac{d}{dt} [a y_1(t) + b y_2(t)] + 3t [a y_1(t) + b y_2(t)] = a t^2 x_1(t) + b t^2 x_2(t)$$

wt. sum of o/p wt. sum of o/p.

$$= t^2 [a x_1(t) + b x_2(t)]$$

weighted sum of i/p.

∴ s/m is linear.

$$2. \frac{dy(t)}{dt} + 2y(t) = x^2(t)$$

$$\text{for } x_1(t), \text{ o/p is } y_1(t).$$

$$\frac{dy_1(t)}{dt} + 2y_1(t) = x_1^2(t)$$

$$(x a) \quad a \frac{dy_1(t)}{dt} + 2a y_1(t) = 2a x_1^2(t) \quad \text{--- ①}$$

Suf for $x_2(t)$, o/p is $y_2(t)$

$$\frac{dy_2(t)}{dt} + 2y_2(t) = x_2^2(t)$$

$$b \frac{dy_2(t)}{dt} + 2by_2(t) = bx_2^2(t) \quad \text{--- (2)}$$

① + ②

$$\Rightarrow a \frac{dy_1(t)}{dt} + b \frac{dy_2(t)}{dt} + 2ay_1(t) + 2by_2(t) = ax_1^2(t) + bx_2^2(t)$$

$$\frac{d}{dt} [ay_1(t) + by_2(t)] + 2[ay_1(t) + by_2(t)] = ax_1^2(t) + bx_2^2(t)$$

wt. sum of o/p wt. sum of o/p.

= $ax_1^2(t) + bx_2^2(t)$

not aft of wt. sum of i/p.

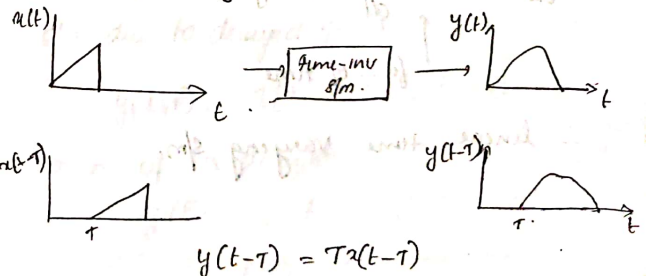
Non-linear s/m, as superposition principle is not satisfied.

→ Time variant and time invariant system

→ A s/m is called time-invariant if its input-output characteristics do not change with time

Suppose $x_1(t)$ is applied to a s/m and $y_1(t)$ is obtained as o/p.

If we delay the i/p by τ sec. then for a time invariant s/m, o/p also will be delayed by τ sec.



1/p delay $y(t, \tau) = \tau[x(t - \tau)]$ $y(t, \tau) = y(t - \tau)$

→ Linear time invariant systems (LTI).

Coefficients of differential equation are constant

→ If coeff. are fn of time then linear time varying sys.

$$2 \frac{d^2 y(t)}{dt^2} + 4 \frac{dy(t)}{dt} + 5y(t) = 5x(t)$$

↑ ↑ ↑ ↑

all coeff are const.

$\therefore LTI$ $q.m.$

$$\frac{d^2 y(t)}{dt^2} + 4t \frac{dy(t)}{dt} + 5y(t) = x(t)$$

$\uparrow$
 fn. of time

$\therefore$ linear time varying s/m.

→ For discrete time systems, check the coefficients of difference equation.

$$y(n) - 4y(n-1) + 3y(n-2) = x(n) \rightarrow \text{LTI}$$

$$y(n) - 3ny(n-1) + n^2 y(n-2) = x(n) + x(n-1)$$

for. of $n \rightarrow$ linear time-varying

→ Determine whether or not the system is time-invariant.

$$y(t) = t x(t)$$

a: $y(t) = \tau[x(t)] = t x(t)$

O/P due to delayed i/P is

$$(1) \quad y(t, \tau) = (\tau[x(t-\tau)]) = t x(t-\tau).$$

when o/p is delayed.

$$y(t-\tau) = (t-\tau)x(t-\tau).$$

$$y(t, \tau) \neq y(t - \tau) \rightarrow \text{time variant.}$$

$$\rightarrow y(t) = e^{x(t)}$$

$$y(t) = T[x(t)] = e^{x(t)}$$

o/p due to delayed i/p.

$$y(t, \tau) = e^{x(t-\tau)}$$

when o/p is delayed.

$$y(t, \tau) = e^{x(t-\tau)}$$

$$y(t, \tau) = y(t-\tau) \rightarrow \text{Time-invariant}$$

$$\rightarrow y(n) = x(2n)$$

$$\text{i/p delayed, } y(n, k) = x(2n-k)$$

$$\text{o/p delay} \rightarrow y(n-k) = x(2(n-k)) = x(2n-2k)$$

$$y(n, k) \neq y(n-k) \rightarrow \text{time variant}$$

$$y(n) = x^2(n-1)$$

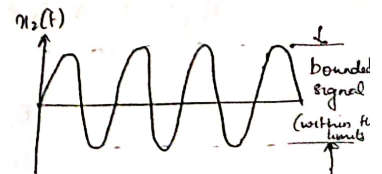
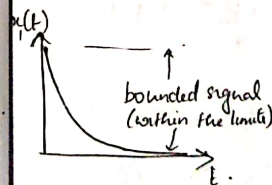
$$y(n, k) = x^2(n-k-1)$$

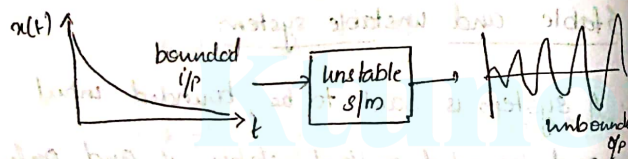
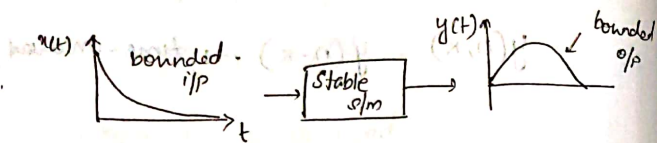
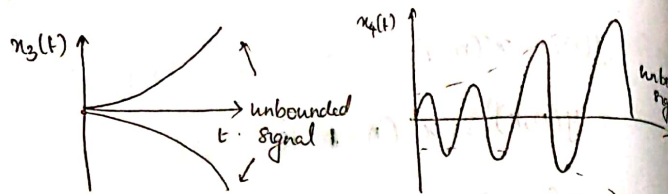
$$y(n-k) = x^2((n-k)-1) = x^2(n-1-k)$$

$$y(n, k) = y(n-k) \rightarrow \text{time-invariant}$$

Stable and unstable system.

A system is said to be bounded input and bounded output stable if and only if every bounded input produces a bounded output.





→ Check whether the following systems are:

1. Static or dynamic
2. Linear or non linear
3. Causal or non causal
4. Time invariant or time variant

$$(i) \quad y(t) \frac{d^2 y(t)}{dt^2} + 3t \frac{dy(t)}{dt} + y(t) = x(t)$$

a: 1) The system is described by differential equation. Hence it is a dynamic s/m

$$2) \quad y_1(t) \frac{d^2 y_1(t)}{dt^2} + 3t \frac{dy_1(t)}{dt} + y_1(t) = x_1(t)$$

$$a \quad y_1(t) \frac{d^2 y_1(t)}{dt^2} + 3at \frac{dy_1(t)}{dt} + ay_1(t) = ax_1(t) - t$$

$$b \quad y_2(t) \frac{d^2 y_2(t)}{dt^2} + 3bt \frac{dy_2(t)}{dt} + by_2(t) = bx_2(t) \quad \rightarrow \textcircled{2}$$

$$\textcircled{1} + \textcircled{2} \Rightarrow$$

$$ay_1(t) + by_2(t) + t[a$$

$$ay_1(t) \frac{d^2 y_1(t)}{dt^2} + 3at \frac{dy_1(t)}{dt} + ay_1(t)$$

$$+ b y_2(t) \frac{d^2 y_2(t)}{dt^2} + 3bt \frac{dy_2(t)}{dt} + by_2(t)$$

$$= a x_1(t) + b x_2(t)$$

$$\frac{d^2}{dt^2} \int ay_1(t)$$

$$\frac{ay_1(t) \frac{d^2 y_1(t)}{dt^2} + by_2(t) \frac{d^2 y_2(t)}{dt^2}}{\text{not weighted sum.}}$$

$$3t \left[a \frac{dy_1(t)}{dt} + b \frac{dy_2(t)}{dt} \right] +$$

$$3t \frac{d}{dt} [ay_1(t) + by_2(t)] \checkmark$$

$$ay_1(t) + by_2(t) \checkmark =$$

$$a x_1(t) + b x_2(t)$$

Does not satisfy superposition

$\therefore$ Non-linear

3. Causal $\rightarrow$ depend only on present values

o/p due to delayed i/p

$$4. y(t, \tau) = x(t - \tau)$$

delayed o/p.

$$y(t - \tau) \frac{d^2 y(t - \tau)}{dt^2} + (3(t - \tau)) \frac{dy(t - \tau)}{dt} + y(t - \tau) = x(t)$$

Coeff. of differential eq. are function of time $\therefore$ Time-variant

$\rightarrow$ Dynamic, non-linear, causal, time-variant

$$\rightarrow y(t) = od\{x(t)\}$$

$$a: y(t) = \frac{1}{2} [x(t) - x(-t)]$$

1. Dynamic or not.

$$y(0) = \frac{1}{2} [x(0) - x(0)]$$

$$y(1) = \frac{1}{2} [x(1) - x(-1)] \rightarrow \text{past}$$

$$y(-1) = \frac{1}{2} [x(-1) - x(1)] \rightarrow \text{future}$$

dynamic

2. Linear or not.

$$y_1(t) = \frac{1}{2} [x_1(t) - x_1(-t)]$$

$$ay_1(t) = \frac{1}{2} [ax_1(t) - ax_1(-t)] \quad \text{--- (1)}$$

$$y_2(t) = \frac{1}{2} [x_2(t) - x_2(-t)]$$

$$by_2(t) = \frac{1}{2} [bx_2(t) - bx_2(-t)] \quad \text{--- (2)}$$

① + ②

$$\Rightarrow ay_1(t) + by_2(t) = \frac{1}{2} [ax_1(t) - ax_1(-t) + bx_2(t) - bx_2(-t)]$$

$$ay_1(t) + by_2(t) = \frac{1}{2} [ax_1(t) + bx_2(t) - \frac{1}{2} [ax_1(-t) + bx_2(-t)]]$$

Linear

3. Causal or non-causal.

$$y(0) = \frac{1}{2} [x(0) - x(0)] \rightarrow \text{present}$$

$$y(1) = \frac{1}{2} [x(1) - x(-1)] \rightarrow \text{present \& past}$$

$$y(-1) = \frac{1}{2} [x(-1) - x(1)] \rightarrow \text{present \& future}$$

$\therefore$ Non-causal.

1. Time-invariant or not.

$$y(t) = \frac{1}{2} [x(t) - x(-t)]$$

o/p due to delayed i/p.

$$y(t, \tau) = \frac{1}{2} [x(t-\tau) - x(-t-\tau)]$$

delayed o/p.

$$y(t-\tau) = \frac{1}{2} [x(t-\tau) - x(-t-\tau)]$$

$$= \frac{1}{2} [x(t-\tau) - x(-t+\tau)]$$

$$y(t, \tau) \neq y(t-\tau)$$

$\therefore$ Time-variant.

$$y(n) = x(n) x(n-1) \quad \text{D, NL, C, TI}$$

$$y(n) = \cos[x(n)] \quad \text{S, NL, C, TI}$$

Representation of LTI s/m

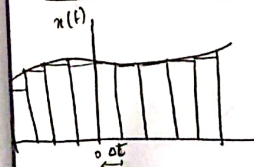
$$a_N \frac{d^N y(t)}{dt^N} + a_{N-1} \frac{d^{N-1} y(t)}{dt^{N-1}} + \dots + a_1 \frac{dy(t)}{dt} + a_0 y(t)$$

$$= b_M \frac{d^M x(t)}{dt^M} + b_{M-1} \frac{d^{M-1} x(t)}{dt^{M-1}} + \dots + b_1 \frac{dx(t)}{dt} + b_0 x(t)$$

For given i/p, o/p of the system can be obtained by solving the differential eq.

o/p can also be obtained by using

convolution integral



$\Delta t \rightarrow 0$ impulse $\delta_\Delta(t) = \frac{1}{\Delta t}$

$$x(0) \delta_\Delta(t) \Delta t, x(\Delta t) \delta_\Delta(t-\Delta t) \Delta t, \dots, x(n\Delta t) \delta_\Delta(t-n\Delta t) \Delta t$$

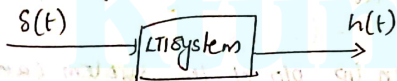
$$x(t) = \sum_{n=-\infty}^{\infty} x(n\Delta t) \delta_\Delta(t-n\Delta t)$$

$$= \int_{-\infty}^{\infty} x(\tau) \delta(t-\tau) d\tau$$

$$x(t) = \int_{-\infty}^{\infty} x(\tau) \delta(t-\tau) d\tau.$$

Convolution integral

if the i/p to the system is an impulse, then the output of the system is denoted by $h(t)$ and is called impulse response of the system.



$$h(t) = \mathcal{T}[\delta(t)]$$

Any signal can be represented as

$$x(t) = \int_{-\infty}^{\infty} x(\tau) \delta(t-\tau) d\tau$$

The system o/p is given by

$$y(t) = \mathcal{T}[x(t)]$$

$$y(t) = \mathcal{T}\left[\int_{-\infty}^{\infty} x(\tau) \delta(t-\tau) d\tau\right]$$

$$y(t) = \int_{-\infty}^{\infty} x(\tau) \mathcal{T}[\delta(t-\tau)] d\tau$$

$$\delta(t) \rightarrow h(t)$$

o/p due to delayed i/p.

$$h(t, \tau) = \mathcal{T}[\delta(t-\tau)]$$

For time invariant s/m,

$$h(t, \tau) = h(t-\tau) = \mathcal{T}[\delta(t-\tau)]$$

$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau \rightarrow \text{Convolution integral}$$

$$y(t) = x(t) * h(t)$$

It is used to find the output of LTI s/m.

Properties of convolution

$$x_1(t) * x_2(t) = \int_{-\infty}^{\infty} x_1(\tau) x_2(t-\tau) d\tau$$

1. Commutative property

$$x_1(t) * x_2(t) = x_2(t) * x_1(t)$$

2. Distributive property

$$x_1(t) * [x_2(t) + x_3(t)] = x_1(t) * x_2(t) + x_1(t) * x_3(t)$$

3. Associative property

$$x_1(t) * [x_2(t) * x_3(t)] = [x_1(t) * x_2(t)] * x_3(t)$$

4. Shift Property

$$\text{If } x_1(t) * x_2(t) = z(t)$$

then,

$$x_1(t) * x_2(t-T) = z(t-T)$$

5. Convolution with an impulse - Unit impulse

$$x(t) * \delta(t) = x(t) \quad \delta(t) = 1 \text{ at } t=0$$

Proof:

$$x(t) * \delta(t) = \int_{-\infty}^{\infty} x(\tau) \delta(t-\tau) d\tau$$

$$= x(t)$$

Impulse response and Convolution sum

$$x(n) \rightarrow \boxed{T} \rightarrow y(n)$$

$$y(n) = T[x(n)]$$

Impulse response.

$$h(n) = T[\delta(n)]$$

Any signal $x(n]$ can be represented as

$$x(n) = \sum_{k=-\infty}^{\infty} x(k) \delta(n-k)$$

$$y(n) = T \left[\sum_{k=-\infty}^{\infty} x(k) \delta(n-k) \right]$$

$$= \sum_{k=-\infty}^{\infty} x(k) T[\delta(n-k)]$$

$\neq$

$$\delta(n) = 1 \text{ at } n=0$$

$$\delta(n-k) = 1 \text{ at } n=k$$

Convolution integral

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) \cdot h(t-\tau) d\tau$$

steps involved in performing calculation

for $x(t)$ $h(t)$

1. replace t by τ

$x(\tau)$ $h(\tau)$
 $\downarrow$ $\downarrow$
 fixed shifted

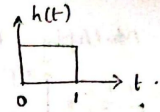
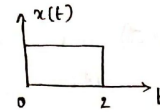
2. Time reversal $h(-\tau)$

3. Time shifting $h(t-\tau)$

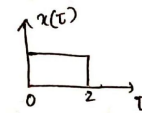
4. Multiply $x(\tau) \cdot h(t-\tau)$. till result = 0.

5. $\int_{-\infty}^{\infty} x(\tau) \cdot h(t-\tau) d\tau$

Ex:



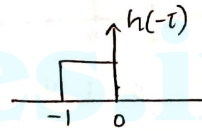
1. $x(\tau)$



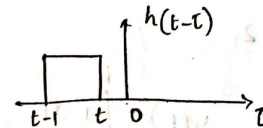
$h(\tau)$



2. Time reversal $h(-\tau)$

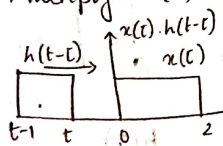


3. Time shifting $h(t-\tau)$



$t \rightarrow -\infty$ to ∞

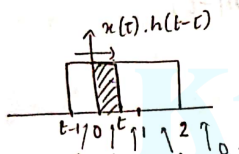
4. Multiply $x(\tau) \cdot h(t-\tau)$



Case 1 : $t < 0$

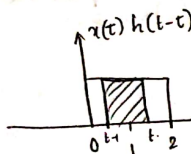
$$y(t) = 0$$

Case 2 : $0 < t < 1$



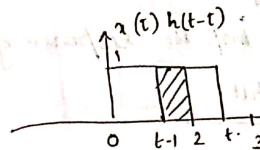
$$y(t) = \int_0^t 1 d\tau = (\tau)_0^t = t$$

Case 3 : $1 < t < 2$



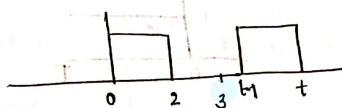
$$y(t) = \int_{t-1}^2 1 d\tau = (\tau)_{t-1}^2 = 2 - t + 1 = 3 - t$$

Case 4 : $2 < t < 3$



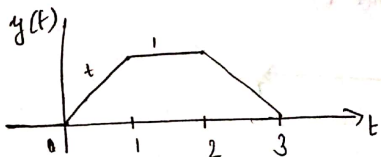
$$y(t) = \int_{t-1}^2 1 d\tau = (\tau)_{t-1}^2 = 2 - t + 1 = 3 - t$$

Case 5 : $t > 3$

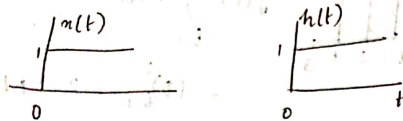


$$y(t) = 0$$

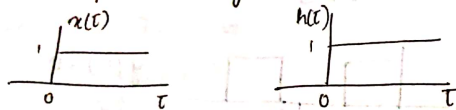
$$y(t) = \begin{cases} 0 & t < 0 \\ t & 0 < t < 1 \\ 1 & 1 < t < 2 \\ 3-t & 2 < t < 3 \\ 0 & t > 3 \end{cases}$$



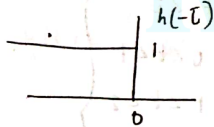
→ Find the output $y(t)$ of an LTI system for the input and impulse response given



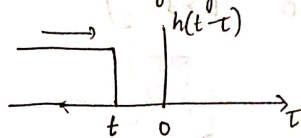
A: 1. Replace t by τ



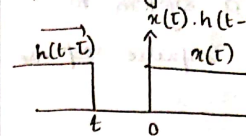
2. Time reversal. $h(-\tau)$



3. Time shifting ($h(t-\tau)$)



4. Multiply $x(\tau) \cdot h(t-\tau)$



Case 1: $t < 0$

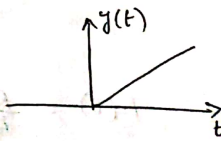
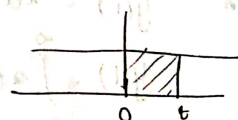
$$y(t) = 0$$

Case 2: $t > 0$

$$y(t) = \int_0^t 1 d\tau$$

$$= (\tau)_0^t = t$$

$$y(t) = \begin{cases} 0 & t < 0 \\ t & t > 0 \end{cases}$$



→ Find the response of a system to unit step input whose impulse response is given by

$$h(t) = \frac{R}{L} e^{-\frac{R}{L}t} u(t)$$

A: $y(t) = x(t) * h(t)$

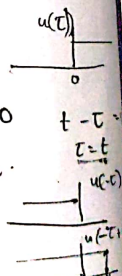
$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau$$

$$x(\tau) = u(\tau) = \begin{cases} 1 & \tau \geq 0 \\ 0 & \tau < 0 \end{cases}$$

$$h(\tau) = \frac{R}{L} e^{-\frac{R}{L}\tau} u(\tau)$$

$$h(t-\tau) = \frac{R}{L} e^{-\frac{R}{L}(t-\tau)} u(t-\tau)$$

$$u(t-\tau) = \begin{cases} 1 & t-\tau \geq 0 \\ 0 & t-\tau < 0 \end{cases} \Rightarrow \begin{cases} \tau \leq t \\ \tau \geq 0 \end{cases} \Rightarrow 0 \leq \tau \leq t$$



$$\begin{aligned} y(t) &= \int_0^t \frac{R}{L} e^{-\frac{R}{L}(t-\tau)} d\tau \\ &= \frac{R}{L} \int_0^t e^{-\frac{R}{L}(t-\tau)} d\tau \\ &= \frac{R}{L} \int_0^t e^{-\frac{R}{L}t} e^{\frac{R}{L}\tau} d\tau \\ &= \frac{R}{L} e^{-\frac{R}{L}t} \int_0^t e^{\frac{R}{L}\tau} d\tau \\ &= \frac{R}{L} e^{-\frac{R}{L}t} \left[e^{\frac{R}{L}\tau} \right]_0^t \cdot \frac{L}{R} \\ &= e^{-\frac{R}{L}t} \left[e^{\frac{R}{L}t} - e^0 \right] \\ &= e^{-\frac{R}{L}t} (e^{\frac{R}{L}t} - 1) \\ &= e^0 - e^{-\frac{R}{L}t} \\ &= 1 - e^{-\frac{R}{L}t} \text{ for } t \geq 0 \end{aligned}$$

→ Find the response of a system in

which $x(t) = e^{-3t} u(t)$

$h(t) = u(t-1)$

n: $y(t) = x(t) * h(t)$

$= \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau$

$x(\tau) = e^{-3\tau} u(\tau)$

$= e^{-3\tau} \quad \tau \geq 0$

$u(\tau) = \begin{cases} 1 & \tau \geq 0 \\ 0 & \tau < 0 \end{cases}$

$h(\tau) = u(\tau-1)$

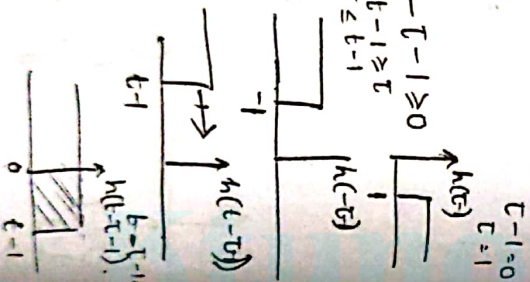
$h(t-\tau) = u(t-\tau-1)$

$\tau \geq 0 \quad \tau \leq t-1 \Rightarrow 0 \leq \tau \leq t-1$

$y(t) = \int_0^{t-1} e^{-3\tau} d\tau$

$= \frac{e^{-3\tau}}{-3} \Big|_0^{t-1}$

$= \frac{e^{-3(t-1)} - 1}{-3}$



$= \frac{1}{3} \left(1 - e^{-3(t-1)} \right)$

→ Evaluate the output $y(t)$ when

$x(t) = e^{-at} u(t) ; a > 0$

and $h(t) = u(t)$

n: $y(t) = x(t) * h(t)$

$= \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau$

$x(\tau) = e^{-a\tau} u(\tau)$

$u(\tau) = \begin{cases} 1 & \tau > 0 \\ 0 & \tau < 0 \end{cases}$

$= e^{-a\tau} \quad \tau > 0$

$h(\tau) = u(\tau)$

$h(t-\tau) = u(t-\tau)$

$h(t-\tau) = u(t-\tau)$

$\tau \geq 0 \quad t-\tau \geq 0 \Rightarrow \tau \leq t$

$\tau > 0 \quad \& \tau \leq t \Rightarrow 0 \leq \tau \leq t$

$y(t) = \int_0^t e^{-a\tau} d\tau = \frac{e^{-a\tau}}{-a} \Big|_0^t = \frac{1 - e^{-at}}{a}$